Diagramme de classes - Gestion reseau

Version simplifiee correspondant au code Java actuel

NetworkManager

- devices: List<NetworkDevice>
- deviceIds: Set<String>
- devicesById: Map<String, NetworkDevice>

+ addDevice(d)
+ removeDevice(id)
+ findDevice(id)
+ getDevicesByStatus(status)
+ discover()
+ raiseAlert(a)

Alert

- channel: String
- severity: String
- message: String

+ send()

NetworkDevice

id: String
ipAddress: String
status: String
protocol: ManagementProtocol

+ reboot(): void (abstract)
+ getInfo(): String (abstract)
+ applyConfig(cfg): void (abstract)
+ backupConfig(): String (abstract)
+ getStatus(): String
+ collectMetrics(): Map

ManagementProtocol

+ open(ip)
+ send(cmd)
+ close()

Router

+ reboot()
+ getInfo()
+ applyConfig(cfg)
+ backupConfig()

Switch

+ reboot()
+ getInfo()
+ applyConfig(cfg)
+ backupConfig()

Firewall

+ reboot()
+ getInfo()
+ applyConfig(cfg)
+ backupConfig()

ProtocolSimulator

- name: String
  (SNMP or SSH)

+ open(ip)
+ send(cmd)
+ close()

Relations filled diamond: device management | dashed: dependency / implementation | triangle: inheritance

Protocols, alerts and metrics are simulated.